What is a differential, really? (Optional)

If the introduction of differentials on the previous page seemed unmotivated and little more than a change of notation, here is a more precise discussion. This is particularly important in multivariable calculus, where the differential is the key to defining the derivative, so maybe review this when you take MVC.

Definition.

(a) A function $L : \mathbb{R} \to \mathbb{R}$ of the form $L(x) = mx$ for some fixed real number m is called *linear*.

(b) A function $A : \mathbb{R} \to \mathbb{R}$ of the form $A(x) = mx + b$ for some fixed real number m is called *affine*.

Note that every linear function is affine (with $b = 0$), but an affine function with $b \neq 0$ is not linear.

If we fix a real number a, then we can write any real number x as $x = a + \underbrace{(x-a)}_{\Delta x} = a + \Delta x$. We see that an affine function $f(x) = mx + b$ has the property that

$$f(x) = f(a+\Delta x) = m(a+\Delta x) + b = (ma+b) + m\Delta x = f(a) + L(\Delta x),$$

where L (as a function of Δx) is linear.

Definition. Let $f : \mathbb{R} \to \mathbb{R}$ be an affine function.

1. For every real number a, the associated linear function L above is called the *differential of f at a*, which we denote by df_a.
2. The *differential of f* is then the function $df : a \mapsto df_a$ which assigns to each real number a the differential of f at a.

If we let $\Delta y = f(x) - f(a)$ denote the change in the value of a function f resulting from changing $a \mapsto a + \Delta x$, then we see from the above that for an affine function $f = mx + b$, $\Delta y = df_a(\Delta x) = m\Delta x$. That is, the change Δy for an affine function is simply equal to the change Δx, magnified or diminished by a factor of m.

We have seen that, if f is differentiable at a, then $f(x) \approx f(a) + f'(a)\Delta x$ and therefore $\Delta y \approx f'(a)\Delta x$. In the language above, if f is differentiable at a, then for x near a f is approximately an affine function whose differential is the linear function which multiplies Δx by $f'(a)$.

Definition. Let f be a differentiable function.

1. We define the *differential of f at x* to be the linear function df_x defined by
$$df_x(\Delta x) = f'(x)\Delta x.$$
2. We define the *differential of f* to be the function which assigns $x \mapsto df_x$ for each real number x in the domain of f'.

Notation. Since Δy is only approximately equal to $df_x(\Delta x)$, to distinguish these we write
$$dy = df_x(\Delta x) = f'(x)\Delta x. \tag{1}$$

If we consider the special case $f(x) = x$, so that $y = x$, then $f'(x) = 1$, so we have
$$dy = df_x(\Delta x) = 1 \cdot \Delta x = \Delta x.$$

Since $y = x$, we can write $dx = dy = \Delta x$. We can therefore write Equation (1) as
$$dy = f'(x)dx. \tag{2}$$